INCH POUND

MIL-C-3098/56E
10 April 1995
SUPERSEDING
MIL-C-3098/56D
23 June 1975

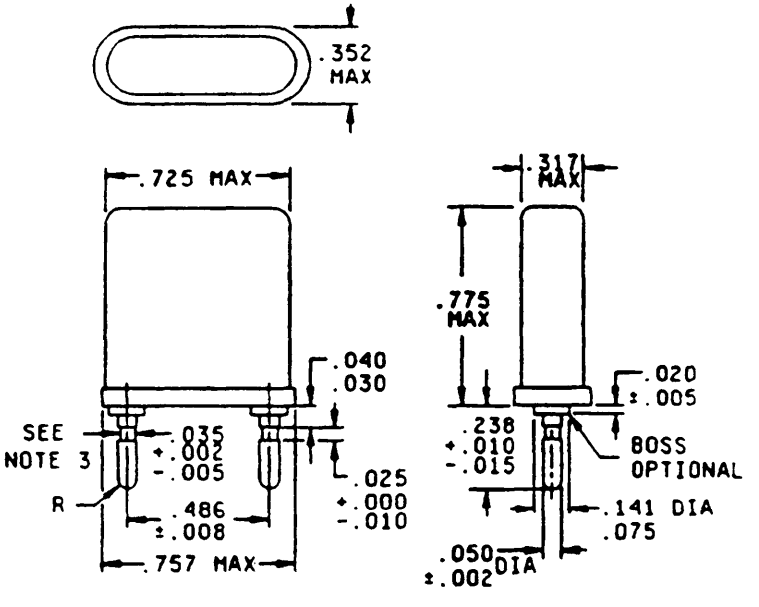
MILITARY SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR-85/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-3098.

Pertinent characteristics: 0.8 MHz to 20 MHz; fundamental; noncontrolled; series.



Inches	mm
.002	0.05
.005	0.13
.008	0.20
.010	0.25
.015	0.38
.020	0.51
.025	0.64
.030	0.76
.035	0.89
.040	1.02
.050	1.27
.075	1.91
.141	3.58
.238	6.05
.317	8.05
.352	8.94
.486	12.34
.725	18.42
.757	19.23
.775	19.69

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The pin undercut may be omitted.
4. Marking shall be in accordance with MIL-C-3098.

FIGURE 1. Crystal unit - CR-85/U.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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REQUIREMENTS:

Holder: HC-6/U (MIL-H-10056/2) or HC-36/U (MIL-H-10056/23) or HC-48/U (MIL-H-10056/35).

Dimensions, marking, and configuration: See figure 1.

Frequency range: 0.8 MHz to 20 MHz, inclusive.

Frequency tolerance:

Primary operating temperature range: ±20 ppm.

Secondary operating temperature range: ±30 ppm.

Equivalent resistance: See table I.

TABLE I. Equivalent resistance.

Frequency range, inclusive	Maximum resistance
MHz	Ohms
0.8 to 0.85	520
0.85+ to 0.9	480
0.9+ to 1	440
1+ to 1.12	400
1.12+ to 1.25	380
1.25+ to 1.37	340
1.37+ to 1.5	300
1.5+ to 1.62	280
1.62+ to 1.75	250
1.75+ to 1.87	220
1.87+ to 2.0	190
2.0+ to 2.12	170
2.12+ to 2.25	150
2.25+ to 2.6	130
2.6+ to 3	90
3+ to 3.4	70
3.4+ to 3.75	52
3.75+ to 4	45
4+ to 5	37
5+ to 7	25
7+ to 10	20
10+ to 15	18
15+ to 20	15

Mode of oscillation: Fundamental.

Operating temperature range:

Primary operating temperature range: -40°C to + 90°C, inclusive.

Secondary operating temperature range: -55°C to -40°C and +90°C to +105°C, inclusive.

Drive level: 0.8 MHz to 10.0 MHz = 10.0 ±2.0 mW.
10.0+ MHz to 20.0 MHz = 5.0 ±1.0 mW.

Capacitance, shunt: 7.0 pf maximum.

Shock (specified pulse):	0.8 MHz to 2.0 MHz	2.0+ MHz to 20.0 MHz
Frequency change permitted:	±10 ppm	±5 ppm
Equivalent resistance change permitted:	±15 percent	±10 percent

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Vibration: MIL-STD-202, method 201.	0.8 MHz to 2.0 MHz	2.0+ MHz to 20.0 MHz
Frequency change permitted:	±10 ppm	±5 ppm
Equivalent resistance change permitted:	±15 percent	±10 percent

Thermal shock:	0.8 MHz to 2.0 MHz	2.0+ MHz to 20.0 MHz
Frequency change permitted:	±10 ppm	±5 ppm
Equivalent resistance change permitted:	±15 percent	±10 percent

Aging:

Frequency change permitted: ±5 ppm.

NOTES:
1. The socket designed to mate with crystal holder HC-6/U is designated in MIL-S-12883/04-01 (see MIL-S-12883/4). For Navy use, other sockets listed in MIL-STD-242 may be used.
2. NATO preferred type NXT55.

Revision letters are not used to denote changes due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI
Navy - AS, MC, SH
Air Force - 19

Preparing activity:
Army - ER

Agent:
DLA - ES

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